Code for making a Randium simulation

This folder contain code for the Randium model of viscous liquid dynamics. The implimentation uses the Numba JIT for efficient calculations on the CPU and GPU.

Setup of Python

Before attempting to running the code, it is recommended to ensure that Numba for works. See Numba's documetation for more: numba.readthedocs.io

Then type following to virtual enviroment:

```
python3 -m venv venv
. venv/bin/activate
pip install -r requirements.txt
```

Run the code with pure functions

The core for running a simulation is in the `backend.py` file. The file can be executed as a script, to conduct a basic simulation:

```
python3 randium_2d_gpu/backend.py
```

Run the code using a front-end class

The file `randium_2d_gpu/randium_2d_gpu.py` contain a class using function in the `backend.py` file. As an usage example, see `run.py`

```
python3 run.sh
```

The output of the simulation is located in the `Data` folder. The result of an simulation is shown in the folder.